EliteDesign: Elite-Guided Learning for RNA Design via the Cross-Entropy Method

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Abstract

The aim of RNA design is to find, within an exponentially large search space, a nucleotide sequence that folds into a given target structure. Efficient RNA design could aid in the development of RNA-based therapeutics such as riboswitches and ribozymes. However, efficiently finding sequences with a high target folding probability remains challenging, particularly for long RNAs. To address this challenge, we introduce EliteDesign, an approach that uses the cross-entropy method to iteratively refine a distribution over valid RNA sequences. Our method outperforms the state-of-the-art RNA design method, SamplingDesign, on Eterna100, achieving a better average target folding probability while being 500 times faster. EliteDesign also scales to much longer RNA structures. On ten 16S rRNA targets ranging from 950 to 1,525 nucleotides, it achieves an average target folding probability of 0.51 versus 0.20 for SAMFEO while reducing the average runtime per puzzle by a factor of 24.